

Lab 6 Report

Second-Order System Analysis of the Quanser QUBE-Servo under Unity Feedback Control

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May 22, 2026

Abstract

This lab focused on analyzing the behavior of a second-order system using the Quanser QUBE-Servo under unity feedback control. The experiment explored important second-order system characteristics such as damping ratio, natural frequency, peak time, and percent overshoot. A Simulink model was developed to implement unity feedback position control, and a step input was applied to observe the servo response. The experimental response was then compared with the theoretical calculations obtained from the closed-loop transfer function. The results demonstrated the relationship between system parameters and transient response behavior, while also reinforcing concepts related to underdamped second-order systems and control system analysis.

1 System Built

The system built in this lab was a unity feedback position control system for the Quanser QUBE-Servo. The setup consisted of the QUBE-Servo hardware connected to a Simulink controller through QUARC. The Simulink model implemented a unity feedback loop where the measured motor position was continuously compared to the desired reference position. The model included a step input, summing junction, gain block, encoder feedback, and motor voltage output. A 1 rad step input was applied at 1 s, and the response was observed over a 2.5 s simulation interval.

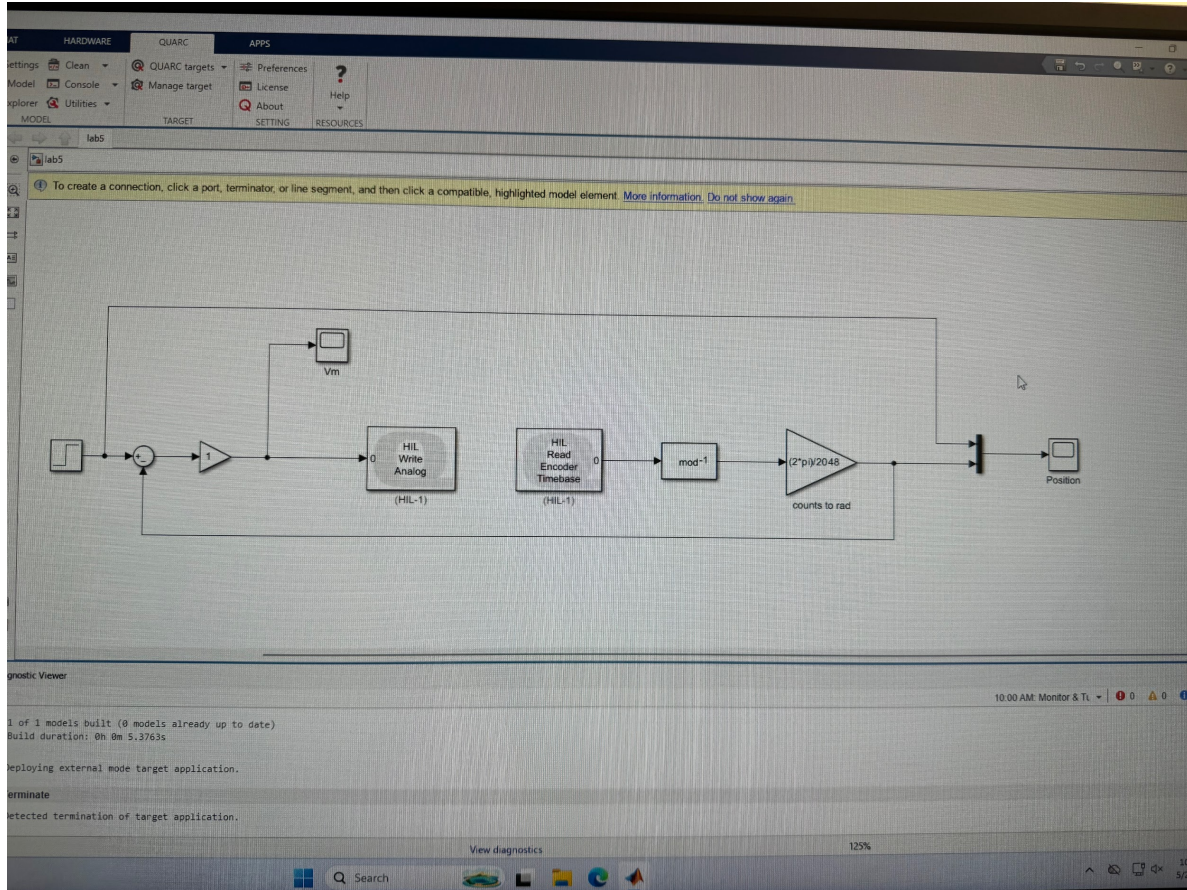


Figure 1: QUARC/Simulink unity feedback position control model for the QUBE-Servo. The loop compares the encoder-measured position (converted from counts to radians) against the reference step and drives the motor through the HIL Write Analog block.

2 In-Lab Exercises and Results

2.1 Natural Frequency and Damping Ratio

Given the QUBE-Servo closed-loop equation under unity feedback (Equation 6 in the lab manual) and the model parameters, the closed-loop response takes the standard second-order form

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \quad (1)$$

Matching the closed-loop denominator to this standard form yields

$$\omega_n = 13.30 \text{ rad/s}, \quad \zeta = 0.29. \quad (2)$$

2.2 Expected Peak Time and Percent Overshoot

For an underdamped second-order system, the peak time and percent overshoot are

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}, \quad \%OS = 100 e^{-\zeta\pi/\sqrt{1 - \zeta^2}}. \quad (3)$$

Substituting the obtained ω_n and ζ gives the expected values

$$t_p \approx 0.25 \text{ s}, \quad \%OS \approx 32\%. \quad (4)$$

2.3 Building and Running the QUARC Controller

The QUARC controller was built and run on the hardware. The resulting scopes matched the expected form shown in the lab manual (Figure 1 shows the model that generated them). Figure 2 shows the QUBE-Servo position response with both the setpoint and the measured position on a single scope, and Figure 3 shows the corresponding motor voltage command.

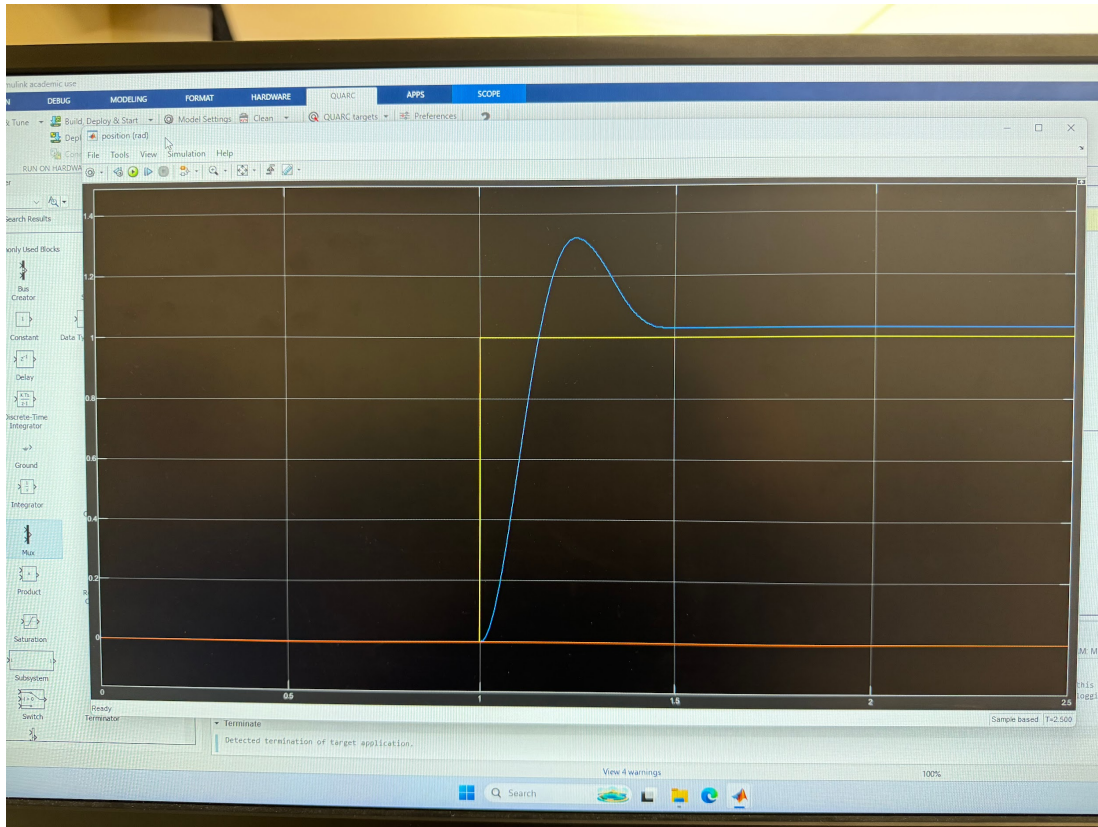


Figure 2: QUBE-Servo position response. The yellow trace is the 1 rad setpoint applied at 1 s; the blue trace is the measured motor position, which overshoots before settling, confirming underdamped behavior.

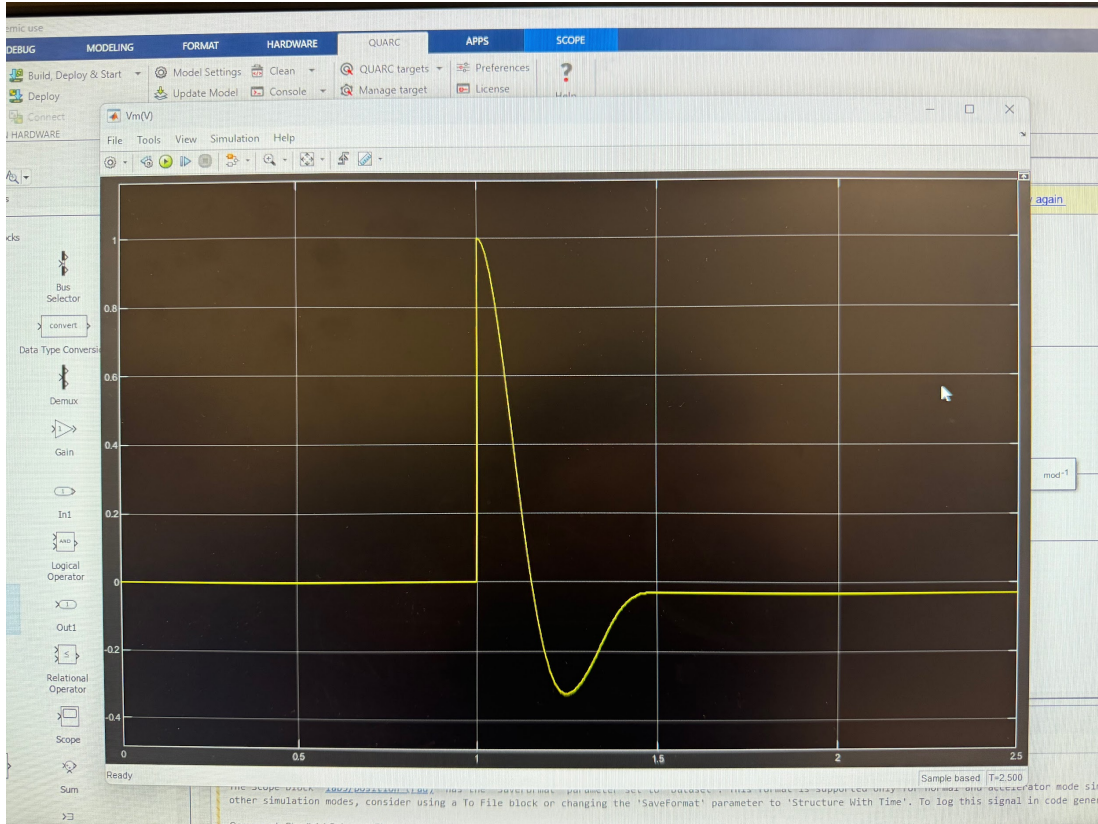


Figure 3: Motor voltage command V_m produced by the unity feedback controller in response to the step input.

2.4 Measured Peak Time and Percent Overshoot

The peak time and percent overshoot were extracted from the logged response using a short MATLAB script (Figure 4). The experimental response reached its maximum value of $y_{\max} = 1.3232$ rad at $t_{\max} = 1.2380$ s. Since the step input occurred at $t_0 = 1.0000$ s, the measured peak time was

$$t_p = t_{\max} - t_0 = 0.2380 \text{ s}, \quad (5)$$

and the measured percent overshoot was $\%OS = 32.2291\%$. These values were close to the theoretical results, with the experimental peak time slightly faster than expected and the overshoot within a fraction of a percent of the predicted 32%.

Table 1: Comparison of theoretical and measured transient-response metrics.

Metric	Theoretical	Measured
Peak time t_p (s)	0.25	0.2380
Percent overshoot $\%OS$	32%	32.2291%

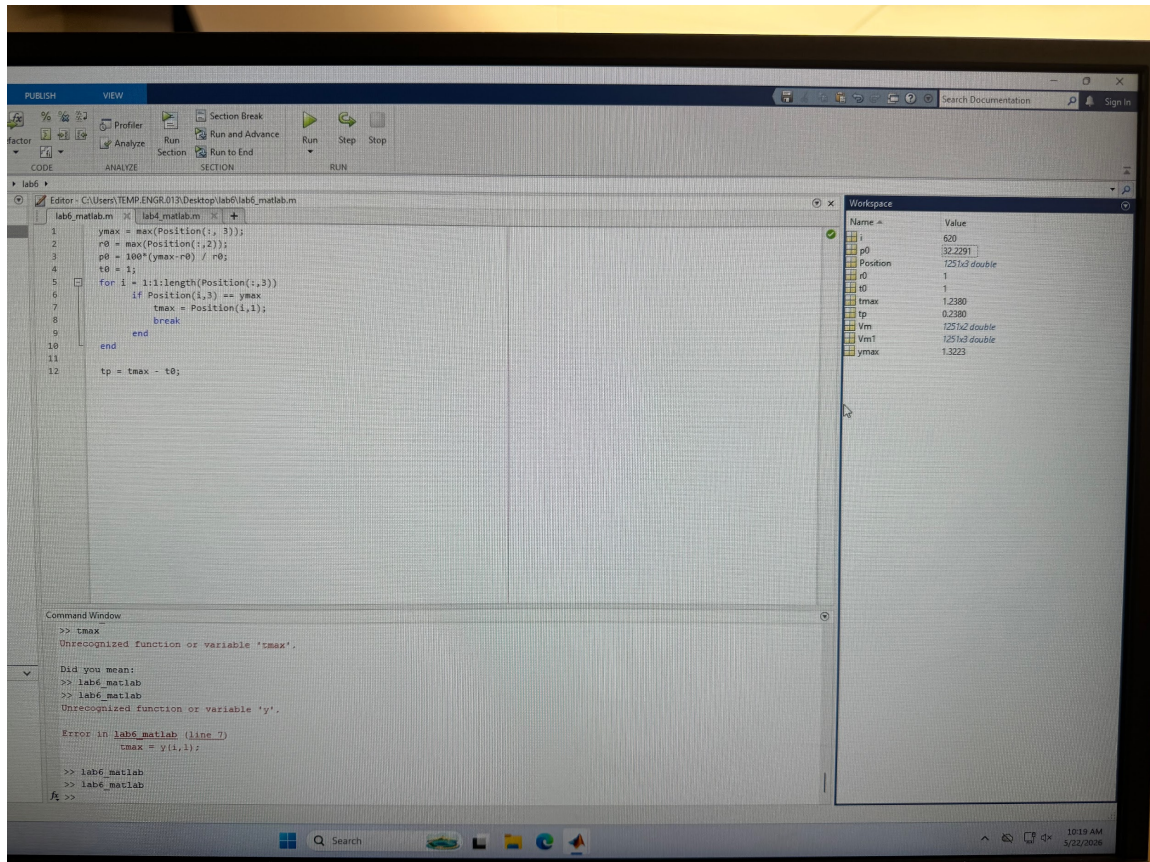


Figure 4: MATLAB script and workspace used to compute the peak value, peak time, and percent overshoot from the logged position data ($y_{\max} = 1.3232$, $t_{\max} = 1.2380$, $t_p = 0.2380$, %OS = 32.2291).

3 Analysis

Results analysis. The results confirmed the expected behavior of an underdamped second-order system. The QUBE-Servo response showed oscillations and overshoot before reaching steady state, which aligned with the calculated damping ratio and natural frequency. The measured peak time and percent overshoot were reasonably close to the theoretical values, demonstrating that the mathematical model accurately represented the system dynamics. The experiment also showed how the damping ratio affects the shape of the response while the natural frequency affects the speed of the response. A lower damping ratio resulted in larger overshoot and oscillation, while the relatively high natural frequency allowed the system to respond quickly.

Issues encountered. The only issue encountered was constructing the code. We worked around it and were able to obtain the correct results.

4 Conclusion

In this lab, a unity feedback position control system was implemented using the Quanser QUBE-Servo to study second-order system behavior. The experiment successfully demonstrated concepts

such as damping ratio, natural frequency, percent overshoot, and peak time. Theoretical calculations were compared with experimental measurements, and the results showed strong agreement between the mathematical model and the physical system response. Overall, the lab provided practical experience with control systems, Simulink modeling, and transient response analysis of underdamped second-order systems.